Simplifying Algebraic Fractions

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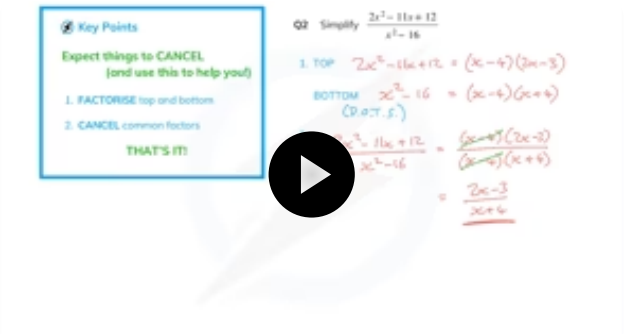

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Simplifying algebraic fractions

What is an algebraic fraction?

- An **algebraic fraction** is a fraction with an **algebraic expression** on the top (numerator) and/or the bottom (denominator)

How do you simplify an algebraic fraction?

- If possible, **factorise fully** the top and bottom
  - E.g. $\frac{2x}{x^2+3x} = \frac{2x}{x(x+3)}$
- **Cancel** common factors
  - This factor may be a **single term**
    - E.g. $\frac{\cancel{x}(5x-1)}{4\cancel{x}} = \frac{5x-1}{4}$
  - It could also be a **common bracket**
    - E.g. $\frac{x\cancel{(x+2)}}{\cancel{(x+2)}(x-1)} = \frac{x}{x-1}$
- A **common mistake** is to cancel a factor that is not common to **all terms** in either the top or the bottom of a

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

- x is not common to **all** terms in the bottom of the fraction
- and the expression on the bottom cannot be factorised

Examiner Tips and Tricks

When asked to **simplify** an algebraic fraction, **factorise** top and bottom.

It is very likely that one of the factors will be the same on the top and the bottom.

Use this fact to help you if one of the expressions is difficult to factorise!

Worked Example

Simplify $\dfrac{4x+6}{2x^2-7x-15}$

Answer:

Factorise the top, by using 2 as a common factor

$$\frac{2(2x+3)}{2x^2-7x-15}$$

Factorise the bottom using your preferred method
Using the fact that the top factorised to $(2x+3)$ may help!

$$\frac{2(2x+3)}{(2x+3)(x-5)}$$

The common factors on the top and bottom reduce to 1 (cancel out)

$$\frac{2\cancel{(2x+3)}}{\cancel{(2x+3)}(x-5)}$$

$$\frac{2}{(x-5)}$$

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